Malhar Gudekar

(217) -766-0681 — gudekar2@illinois.edu — LinkedIn — Portfolio — GitHub

Education

University of Illinois Urbana Champaign August 2024 - May 2026
Master of Science, Information Management GPA: 3.8/4
• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai June 2019 - May 2023
Bachelor of Engineering, Computer Science GPA: 9.25/10
• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

Skills

• **Languages & DSA:** Python, SQL, C++ (basic), Data Structures & Algorithms (Arrays, Trees, Graphs, DP), LeetCode
• **Machine Learning & AI:** PyTorch, Neural Networks, Generative AI, Model Optimization, Inference, NLP, Computer Vision,TensorFlow, scikit-learn, Anomaly Detection, Model Monitoring
• **Systems & Tools:** FastAPI, Distributed Systems, ETL Pipelines, Kafka, PySpark, PostgreSQL, Neo4j, AWS, Docker, Git

Work Experience

Business Intelligence Group August 2025 - December 2025
Technical Consultant Champaign, USA
- Built and deployed scalable ML system using PyTorch, FastAPI, PostgreSQL, and vector embeddings for multi-modal data processing and high-performance production inference
- Designed efficient, fault-tolerant **ETL, async pipelines** to process, index large document datasets for optimized low-latency retrieval systems
- Improved system performance by **64%** through optimized indexing strategies and pipeline-level tuning in distributed production environments
- Developed backend services enabling real-time, high-performance **AI model serving** of unstructured data across scalable distributed systems

University of Illinois Urbana Champaign May 2025 - Present
Research Assistant - iSchool Champaign, USA
- Owned end-to-end design, development, and deployment of cross-platform ML backend system ingesting wearable sensor data for 100+ users, enabling scalable collection and real-time analytics
- Designed and optimized PostgreSQL schema and ML inference pipeline architecture for high-volume sensor datasets, improving query performance by 38% in production workloads
- Built production-ready Python generative AI pipeline using PyTorch and transformer models with retrieval and context summarization, improving response accuracy by 43%
- Developed asynchronous model inference workflows with monitoring and alerting for real-time, fault-tolerant processing of sensor data with focus on production reliability

Research Assistant - CHI AI in Infodemic Management January 2025 - May 2025
- Designed and deployed scalable NLP and anomaly detection pipelines in Python to process high-volume datasets, reducing processing latency by 41% and improving model accuracy
- Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing across scalable systems
- Designed and developed **production-grade REST APIs using FastAPI** to serve AI models, implementing validation, monitoring, and automated deployment workflows for reliable integration
- Enabled large-scale data processing across social media streams using **distributed systems and large-scale processing** principles for optimized performance and scalability

Swift Mobil Software Solutions Provider July 2023 - October 2023
Data Analyst Mumbai, IN
- Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving operational visibility
- Processed and optimized large-scale datasets using **Python** and **PySpark** to identify failure patterns and bottlenecks in distributed systems
- Developed **SQL** and **graph queries** to map dependencies, analyze operations, detect data issues, accelerating root-cause analysis by **26%**

Pscope Technologies Pvt. Ltd. January 2023 - June 2023
Data Analyst Mumbai, IN
- Developed and maintained **20+ Power BI** dashboards across Sales, Finance, and Operations for enterprise clients, leveraging **SQL** and **DAX**, enabling KPI tracking and insights
- Automated recurring **Excel** workflows using **VBA**, implementing validation and scheduling logic, reducing report preparation time by **42%**

Stu/Dio at Illinois August 2025 - Present
Project Manager Champaign, USA
- Led cross-functional team of **6** engineers and designers to deliver **AI-driven product features** for DeepCover game, leveraging feedback and performance insights for prioritization
- Tracked **sprint** progress, delivery metrics, and **QA issues** through **JIRA**-based workflows, managing backlog and resolution for timely delivery

Projects

Scalable ML System for Customer Risk Prediction March 2025
UIUC Hackathon
- Built end-to-end ML pipeline using Python, PyTorch, and LightGBM to classify patterns from large-scale time-series datasets, applying model validation and benchmarking techniques
- Designed and implemented **scalable ETL and feature engineering pipelines**, capturing behavioral trends and temporal patterns such as rolling averages and spending frequency
- Developed a **decision framework for credit line adjustments**, incorporating model outputs and risk thresholds to improve prediction precision